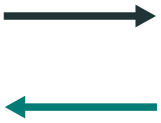


Each route can reach this exit.

Join the steps for this exit.

Back at the start. The exit stays dark.

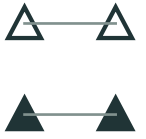
Start with the same blank mark.

Give each route its own mark.

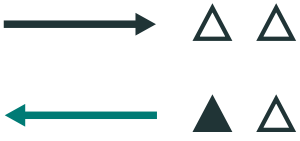
Now this exit can click.

The detector holds the mark.

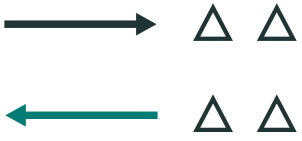
A reader copies it.



In Everett’s account, each reader remembers the matching mark.



Try again. Reverse the copying.



Undo the route marks too.

Join the steps again. Back to dark.

Two routes, one exit, and a record

Reading the marks

A horizontal stroke depicts a signed contribution to the chosen exit. Two equal, opposite steps can cancel. The vertical stroke depicts the exit detector and rays depict a possible click. Open and solid triangles denote distinct record values. In paired columns the left triangle is the detector memory and the right triangle is the reader memory. The open triangle also serves as the ready value. Each strip starts a fresh run. The link between matching marks in strip 3 pairs a reader with a detector.

1 A dark exit

Prepare $|+\rangle = (|\mathrm{L}\rangle + |\mathrm{R}\rangle)/\sqrt{2}$. The recombiner tests $|-\rangle = (|\mathrm{L}\rangle - |\mathrm{R}\rangle)/\sqrt{2}$ at the drawn exit. The two contributions to its amplitude are $+1/2$ and $-1/2$. Their sum is zero. A particle prepared in either route alone gives a click probability of 1/2. Here and below the Born rule and these input and output states are supplied.

2 Give the routes distinct records

A reversible controlled flip maps $|\mathrm{L},0\rangle$ to $|\mathrm{L},0\rangle$ and $|\mathrm{R},0\rangle$ to $|\mathrm{R},1\rangle$. In the coherent superposition the dark-exit record vector becomes $(|0\rangle - |1\rangle)/2$. Distinct ideal records are orthogonal, so its squared length is 1/2. Each joint event (dark exit, specified record) has probability 1/4. Either mark identifies the corresponding route at the tagging interaction.

3 Read the record

Copy the detector bit into an initially ready reader. The joint state is $(|\mathrm{L},0,0\rangle + |\mathrm{R},1,1\rangle)/\sqrt{2}$. Relative to reader value 0 the detector reads 0, and relative to reader value 1 it reads 1. Each conditional state has one matching record. Everett interprets both terms as continuing histories of the whole. In a macroscopic account, persistent copies in an environment help stabilize the records.

4 Reverse the record before testing the exit

On a fresh run, reverse the reader copy and then the detector flip. Each is its own inverse. The coherent state returns to $|+\rangle|0,0\rangle$, so the exit probability returns to zero. A joint classical mixture of the two recorded alternatives instead gives 1/2 after the same reversal. This contrast requires coherent control of every copy, including any environmental record.